

1 *Drosophila melanogaster* is a species of fruit fly.

Fig. 1.1 shows a female fruit fly and a male fruit fly. A typical fruit fly is 3 mm in length.



Fig. 1.1

Scientists have studied the inheritance patterns of many genetic traits in fruit flies.

To carry out genetic crosses with fruit flies:

- a specimen tube is prepared with food for the fruit flies, as shown in Fig. 1.2
- adult male and female fruit flies are added to the specimen tube to allow mating to take place
- the specimen tube is kept in warm conditions for several days
- the eggs laid by female fruit flies develop into pupae
- adults are removed from the specimen tube before pupae mature into adult fruit flies
- offspring emerge as adult flies 10-15 days after eggs are laid.

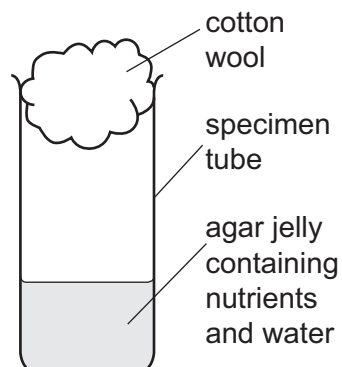


Fig. 1.2

One of the genetic traits studied in fruit flies is eye colour. The normal (wild type) eye colour of *D. melanogaster* is red.

Eye colour in *D. melanogaster* is controlled by several genes, including two genes that are located on separate chromosomes, **A/a** and **D/d**.

- Allele **A** is dominant to allele **a**.
- Allele **D** is dominant to allele **d**.

Table 1.1 summarises eye colour in *D. melanogaster* for these two genes.

Table 1.1

eye colour phenotype	genotypes
red	AADD, AaDD, AaDd, AADd
brown	AAdd, Aadd
scarlet (bright red)	aaDD, aaDd
white	aadd

(a) A student was provided with two populations of fruit fly:

- brown-eyed fruit flies with genotype AAdd
- scarlet-eyed fruit flies with genotype aaDD.

In each population, males and females were provided in separate specimen tubes.

The student decided to carry out two genetic crosses.

The first cross used fruit flies from the initial populations to produce offspring that are heterozygous for each of the two genes (double heterozygotes).

The second cross used the double heterozygotes produced from the first cross.

To carry out the genetic crosses, the student was provided with standard laboratory equipment and:

- specimen tubes containing food
- a chemical to anaesthetise the flies – this chemical, when given at a particular dose, makes the flies immobile for more than 30 minutes
- small brushes for sorting immobile flies without harming them.

(i) Identify a hazard in this investigation **and** state a risk associated with the hazard **and** state **one** precaution that the student should take.

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(ii) Describe a method the student could use to carry out:

- The description of your method should be set out in a logical way and be detailed enough for another person to follow.

[7]



- (iii) Predict the ratio of offspring phenotypes from the cross between parents that are heterozygous for the two genes (AaDd).

You may use this space for any working.

ratio

phenotypes.....

[2]

- (b) After crossing double heterozygotes (genotype AaDd), the student recorded the numbers of offspring in each of the four phenotypic groups.

The student used a chi-squared (χ^2) test to analyse these data.

The null hypothesis for this χ^2 test was:

There is no difference between the expected and observed numbers of offspring in each phenotypic group.

The calculated value of χ^2 was **4.798**.

The student compared **4.798** to the values in Table 1.2.

Table 1.2

degrees of freedom	probability level (p)		
	0.10	0.05	0.01
1	2.706	3.841	6.635
2	4.605	5.991	9.210
3	6.251	7.815	11.345
4	7.779	9.488	13.277
5	9.236	11.070	15.086

Using Table 1.2 and the calculated value of χ^2 of **4.798**, state **and** explain what the student can conclude about the results.

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..... [3]

[Total:13]